# Demonstration of Proposed Features

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| **Date** | 15 March 2024 |
| **Team ID** | XXXXXX |
| **Project Name** | Credit Card Approval Prediction |
| **Maximum Marks** | 1 Mark |

**Demonstration of Proposed Features:**

This document captures all the features that were proposed during the project planning phase and tracks whether each feature was successfully implemented and demonstrated. It serves as evidence of the team's ability to deliver on the proposed Credit Card Approval Prediction solution, which automates the credit card approval decision using machine learning, evaluating applicant financial and demographic data to predict approval or rejection just as real banks do.

| S.No | Feature Name | Description | Status (Implemented / Partial / Pending) | Demonstrated (Yes / No) | Remarks |
|---|---|---|---|---|---|
| 1 | Data Preprocessing & Feature Engineering | Cleaned historical applicant data and engineered features, including converting multi-class payment status codes into binary risk labels for classification | Implemented | Yes | Verified with sample applicant records |
| 2 | Multi-Model Training & Comparison | Trained and compared four classification algorithms: Logistic Regression, Random Forest, XGBoost (Gradient Boosting), and Decision Tree | Implemented | Yes | Accuracy, precision, and recall compared across models |
| 3 | Best Model Selection & Serialization | Identified the best-performing model based on evaluation metrics and saved it for integration into the web application | Implemented | Yes | Model file successfully loaded at runtime |
| 4 | Flask Web Application for Prediction | Built an intuitive web interface where users can enter applicant financial and demographic details to get an instant approval/rejection prediction | Implemented | Yes | Tested with multiple applicant profiles |
| 5 | IBM Watson Machine Learning Deployment | Deployed the trained model to IBM Watson Machine Learning for scalable, real-time, cloud-hosted predictions | Implemented | Yes | Cloud endpoint tested for real-time response |
| 6 | Applicant Risk & Eligibility Scenarios | Supports credit analyst screening, compliance batch review of high-risk applicants, and customer self-service eligibility checks | Implemented | Yes | All three usage scenarios demonstrated successfully |

**Feature Implementation Summary:**

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| **Total Features Proposed** | 6 |
| **Total Features Implemented** | 6 |

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| **Total Features Demonstrated** | 6 |
| **Overall Implementation Rate (%)** | 100% |